HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

DATA VISUALIZATION

LAB 1 REPORT

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Group Members

No.	Full Name	Student ID	Contribution
1	Nguyễn Đỗ Bảo	23127026	20%
2	Lê Nguyên Thảo	23127118	20%
3	Đinh Đại Vũ	23127144	20%
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Table 1: Group members and stated contribution

Task Distribution and Completion Status

No.	Assigned Work	Responsible	Student ID	Completion
1	Data cleaning, preprocessing, and preparation of the final analytical dataset	Nguyễn Thanh Owen	23127447	100%
2	Relational database design and Power BI data-model structure	Đinh Đại Vũ	23127144	100%
3	Objectives 1.1–1.2: Sales performance by category and time	Đinh Đại Vũ	23127144	100%
4	Objectives 2.1–2.2: Finance Structure	Vũ Hoàng Minh	23127427	100%
5	Objectives 3.1–3.2: Product influence analysis and dashboard support	Lê Nguyên Thảo	23127118	100%
6	Objectives 4.1–4.2: Customer ratings and review-signal analysis	Nguyễn Thanh Owen	23127447	100%
7	Objectives 5.1–5.2: Machine learning support and model interpretation	Nguyễn Đỗ Bảo	23127026	100%
8	Power BI dashboard design, layout, and visual organization	Lê Nguyên Thảo	23127118	100%
9	Report writing, editing, and final content review	Vũ Hoàng Minh	23127427	100%
10	Report writing, editing, and final content review	Nguyễn Đỗ Bảo	23127026	100%

Table 2: Task allocation used in this report

1 Introduction to the Analytical Problems

1.1 Domain Situation

At the beginning of the project, our group considered **Shopee** because it is a familiar and important platform in Vietnam. However, during the early collection stage, the crawling workflow on Shopee was not stable enough for a reproducible pipeline. The platform structure created practical difficulties for repeated extraction, which would have reduced the consistency of the later analysis. Our group therefore switched to **Lazada** [1], where product and review collection could be implemented more reliably through a hybrid approach that combines browser automation with API-style requests.

Within Lazada, our group selected the **Nike Flagship Store** as the case study. This choice gives the project a focused but still information-rich analytical setting. Because the store belongs to a single major brand, the dataset avoids excessive cross-shop noise and makes interpretation more coherent. At the same time, the store still contains a diverse catalog of footwear, apparel, and accessories across multiple categories and price levels, together with substantial review activity. As a result, the final dataset is suitable for our common analytical problem: examining how category structure, pricing, discount policy, stock availability, ratings, and review behavior influence the sales performance of Nike products on Lazada during the observed period.

1.2 Common Analytical Problem

Based on the collected dataset, our group defines the common analytical problem of the project as follows:

This study investigates how product category, pricing structure, discount policy, stock availability, consumer ratings, and review behavior are associated with the sales performance of Nike products listed on the Lazada platform over the observational period spanning February 20, 2026 to March 20, 2026.

This common analytical problem is appropriate for the project for three reasons. First, it is directly built from the variables available in our self-collected dataset, so the analysis can be supported by actual evidence rather than external assumptions. Second, it connects all analytical themes in the report, including category performance, pricing structure, product influence, customer review signals, and the optional machine learning component. Third, it remains closely related to sales activity on an e-commerce platform, which is the central requirement of the lab.

1.3 Analytical Objectives

1.3.1 Sales Performance by Category and Time

- **Objective 1**: Measure how total revenue is distributed across Nike product categories from February 20, 2026 to March 20, 2026 in order to identify the highest-contributing categories and assess whether sales performance is concentrated in a small number of categories.
  - **Specific**: Category-level revenue contribution analysis aligned with the KPI cards and the `Total Revenue by Category name` bar chart.

- **Measurable**: Uses `Total Daily Revenue`, `Average Revenue per Day`, `Revenue Difference`, and `Total Revenue by category_name`.
- **Achievable**: All required revenue measures and category fields are available in the prepared sales dataset and current dashboard design.
- **Relevant**: Directly supports the group's analytical problem by showing how product category influences sales performance.
- **Time-bound**: Evaluated within the analytical window from 20-02-2026 to 20-03-2026.

- **Objective 2**: Compare the most volatile and most stable Nike product categories over February 20, 2026 to March 20, 2026 in order to determine which categories show consistent sales performance over time and which categories experience stronger revenue fluctuations.
  - **Specific**: Time-based category stability analysis using the `Top 3 Most Volatile Categories` and `Top 3 Most Stable Categories` visuals.
  - **Measurable**: Uses `Category Revenue CV` to compare the relative revenue fluctuation of categories across the analytical period.
  - **Achievable**: The required category-level revenue variation metric is available in the prepared model outputs and dashboard visuals.
  - **Relevant**: Directly supports the analytical problem by showing how category-level sales performance changes over time, not just in total size.
  - **Time-bound**: Evaluated within the same analytical window from 20-02-2026 to 20-03-2026.

1.3.2 Finance Structure

- **Objective 1**: Measure revenue-share differences across predefined Price Band groups from February 20, 2026 to March 20, 2026 in order to identify the dominant price band in the current dashboard design.
  - **Specific**: Price-band contribution analysis aligned with the `Price Distribution` visual.
  - **Measurable**: Uses share of `daily_revenue` and `average selling_price` by Price Band.
  - **Achievable**: `selling_price` and sales variables are available in the prepared snapshot dataset.
  - **Relevant**: Directly supports interpretation of the KPI cards and pie-chart structure on the dashboard.
  - **Time-bound**: Evaluated within the analytical window from 20-02-2026 to 20-03-2026.
- **Objective 2**: Use the `Top Trending` line chart together with KPI cards and dashboard slicers from February 20, 2026 to March 20, 2026 in order to determine whether sales

performance is broad-based or concentrated in a small number of dates or product groups.

- **Specific**: Dashboard-based sensitivity analysis using the `Top Trending` line chart, KPI cards, and slicers.
- **Measurable**: Uses changes in `Total Daily Revenue`, `Average Daily Units Sold`, and `Average Price per Product` under selected date and filter states.
- **Achievable**: The required KPIs, segment dimensions, and filters are available in the current dashboard and source tables.
- **Relevant**: Matches the interactive dashboard workflow and provides visual evidence for report/viva interpretation.
- **Time-bound**: Limited to the same analytical period from 20-02-2026 to 20-03-2026.

1.3.3 Product Influence Analysis

- **Objective 1**: Identify the top Nike products driving unit performance from February 20, 2026 to March 20, 2026 using the `Top Products by Average Unit Sold` visual and dashboard filters.
  - **Specific**: Product-level comparison focused on the leading products by unit performance.
  - **Measurable**: Uses ranking of products by the `Avg Daily Units Sold` measure shown in the current visual.
  - **Achievable**: `Product name`, unit measures, and supporting product attributes are available in the current dashboard model.
  - **Relevant**: Directly supports the page purpose of understanding product influence on sales outcomes.
  - **Time-bound**: Evaluated within the analysis window from 20-02-2026 to 20-03-2026.
- **Objective 2**: Analyze time-based spikes and concentration risk among top products from February 20, 2026 to March 20, 2026 by combining the `Top Unit` trend visual with the dashboard slicers.
  - **Specific**: Temporal sensitivity analysis of top-product unit performance under interactive filters.
  - **Measurable**: Uses peak values of the `Avg Daily Units Sold` trend by date, spike frequency of top products, and comparison of product-level peaks within the selected period.
  - **Achievable**: `Date`, `product`, and unit metrics are available in the current dashboard visuals.
  - **Relevant**: Explains whether page-level performance is broad-based or overly dependent on a small number of products.

- **Time-bound**: Limited to the same analytical period from 20-02-2026 to 20-03-2026.

1.3.4 Customer Ratings and Review Signals

- **Objective 1**: Use the `Monthly Rating and Review Trends` visual together with the year and category slicers to examine how monthly review count and average rating change within a selected year, in order to identify the periods in which the selected Nike categories receive the strongest customer evaluation.
  - **Specific**: Monthly analysis of `review_count` and `average_rating` for the selected year and category context.
  - **Measurable**: Uses monthly values of `review_count` and `average_rating` shown in the `Monthly Rating and Review Trends` visual.
  - **Achievable**: The required review date, category, and rating fields are available in the prepared review dataset and current dashboard design.
  - **Relevant**: Supports the analytical problem by showing when customer satisfaction signals are strongest within the observed review period.
  - **Time-bound**: Evaluated within a selected year from the observed review period 2024 to 2026, with month-level comparison inside that year.
- **Objective 2**: Use the `Monthly Comment Trends for Top Products` line chart together with the year and category slicers to compare monthly comment volume among the most-discussed Nike products within a selected year, in order to determine which products consistently attract the most customer attention and whether review activity is concentrated in a small number of products.
  - **Specific**: Product-level monthly comparison of comment volume for the most-commented products in the selected year.
  - **Measurable**: Uses monthly `review_count` or comment count by `product_name` in the `Monthly Comment Trends for Top Products` visual.
  - **Achievable**: The required product identifiers, review dates, and comment counts are available in the prepared review dataset and dashboard visuals.
  - **Relevant**: Supports the analytical problem by showing which products generate the strongest customer discussion and how that attention is distributed across products.
  - **Time-bound**: Evaluated within a selected year from the observed review period 2024 to 2026, with month-level comparison inside that year.

1.3.5 Machine Learning Insights (Clustering)

- **Objective 1: Clustering Quality and Segment Identification** - Use the silhouette score comparison bar chart and the `Product Segments by Units Sold vs Revenue` scatter chart to evaluate the clustering solution and identify distinct Nike product

groups from February 20, 2026 to March 20, 2026, in order to show that the dataset can be meaningfully segmented into different sales-performance patterns.

  - **Specific**: Compare clustering quality across candidate cluster settings using `silhouette_score`, then visualize product distribution by `segment_id` on the `Product Segments by Units Sold vs Revenue` scatter chart.
  - **Measurable**: Uses differences in `silhouette_score` across cluster options and visible separation of products by `segment_id`, `avg_daily_units_sold`, and `avg_daily_revenue` on the scatter plot.
  - **Achievable**: All required fields are available in `segmentation_metrics.csv` and `product_segments.csv`.
  - **Relevant**: Supports the analytical problem by showing that products can be grouped into meaningful clusters before interpreting sales behavior.
  - **Time-bound**: Evaluated within the analysis window from 20-02-2026 to 20-03-2026.

- **Objective 2: Cluster Revenue Contribution Interpretation** - Use the `Product Segments by Units Sold vs Revenue` scatter chart together with the pie chart of revenue contribution by cluster to interpret how each product segment contributes to overall Nike sales performance from February 20, 2026 to March 20, 2026, in order to explain which clusters dominate total revenue and which clusters represent smaller but distinct performance groups.

  - **Specific**: Compare revenue contribution across `segment_id` and relate each cluster's share of revenue to its position in the `Product Segments by Units Sold vs Revenue` scatter chart.
  - **Measurable**: Uses % revenue contribution by cluster, cluster membership on the scatter chart, and differences in `avg_daily_units_sold` and `avg_daily_revenue`.
  - **Achievable**: The required fields are available from `product_segments.csv` and `segment_profiles.csv`.
  - **Relevant**: Directly supports the analytical problem by linking clustered product characteristics to overall sales performance contribution.
  - **Time-bound**: Limited to the same analytical period from 20-02-2026 to 20-03-2026.

2 Data Collection, Construction, and Preprocessing

2.1 Product Collection

Note: The extraction procedures described in this section were programmatically implemented and executed via the ***products.py*** script.

2.1.1 Objectives and Approach

- **Objective**: To systematically collect all product metadata from the Nike Flagship Store on Lazada. The target data fields include product name, selling price, quantity sold, rating score, and inventory status.
- **Approach**: The system utilizes a Hybrid Web Scraping methodology. This approach combines browser automation (via Selenium) to bypass initial security mechanisms such as Captchas with direct HTTP requests to optimize extraction speed and efficiency by querying hidden internal APIs.

2.1.2 Detailed Execution Steps

- **Step 1: Session Initialization & Authentication**
  - The system initializes an automated browser instance (Google Chrome) using **Selenium** and **WebDriverManager**.
  - It navigates to the target store's homepage. At this point, the execution thread is intentionally paused to allow for manual human intervention. This is necessary to solve complex Captchas or perform account authentication if required by the platform.
  - Upon successful manual authentication, the system extracts the session **Cookies** and the browser's **User-Agent** string.
  - The automated browser is then terminated to free up system memory, and the process transitions to a lightweight HTTP request mode.
- **Step 2: Category Discovery (Structure Extraction)**
  - Utilizing the extracted Cookies and User-Agent, the system simulates an asynchronous HTTP request (AJAX/XMLHttpRequest) directed at the platform's filter API.
  - The server responds with a JSON payload containing the complete categorical architecture of the store.
  - The system parses this JSON to isolate the list of product categories and their corresponding internal identifier codes (values), which serve as inputs for the next phase.
- **Step 3: Data Extraction**
  - **Automated Pagination**: The system iterates through each discovered category. Within a category, a while loop is employed to navigate through sequential product

pages (page 1, page 2, etc.) by dynamically modifying the page parameter in the API URL.

  - **Data Parsing**: For each queried page, the system receives a JSON array of products. The parser extracts core variables: *Item ID*, *Product Name*, *Selling Price*, *Original Price*, *Rating Score*, *Review Count*, *Warehouse Location*, *Stock Status*, *Brand Name*, and *the Product URL*.
  - **Data Cleaning**: Unstandardized data formats are processed immediately. For instance, text strings representing sales volume (e.g., "1.5k sold") are converted into standard integer formats (1500) using regular expressions to facilitate downstream quantitative analysis.
  - **Anti-bot Evasion Mechanism**: The script implements randomized delays using `time.sleep`. Each pause lasts about 2 to 3 seconds between page requests. This simulates organic human browsing behavior and helps reduce the risk of IP blocking or rate-limiting by the platform's anti-scraping firewalls.
  - The pagination loop terminates when the API returns an empty product list, indicating the end of the category.

- **Step 4: Data Organization and Storage**
  - The system dynamically generates a hierarchical directory structure based on the current execution date (`data/MM-DD-YYYY`).
  - The data is simultaneously exported into two formats:
    * **CSV File** (`products.csv`): Intended for end-users to easily read, filter, and visualize the data using spreadsheet software like Microsoft Excel.
    * **JSON File** (`products.json`): Preserves the raw structured format, acting as the requisite input file for the subsequent phase of the project (Review Collection Process).

2.2 Review Collection

Note: The extraction procedures described in this section were programmatically implemented and executed via the ***reviews.py*** script.

2.2.1 Objectives and Approach

- **Objective**: To systematically extract comprehensive customer feedback, including text reviews, star ratings, and timestamps, for all previously identified products to facilitate product quality analysis.
- **Approach**: The system employs an API reverse-engineering technique targeting the platform's MTOP architecture. This methodology relies on dynamic cryptographic token generation (MD5 signing), coupled with a local checkpoint system and automated Captcha handling via Selenium, ensuring continuous and resilient data extraction.

2.2.2 Detailed Execution Steps

- **Step 1: Data Initialization and Checkpointing**
  - The system parses the *products.json* file generated in the first phase. To optimize execution time and prevent data duplication in the event of network interruptions, it implements a checkpoint mechanism. It compares the input list against the existing *reviews.json* output file to bypass products that have already been successfully scraped.
- **Step 2: Session Initialization and Token Extraction**
  - A Selenium-driven browser is temporarily launched to navigate to a sample product page. The primary purpose of this action is not to parse the HTML document, but to force the e-commerce server to issue a specific session cookie (_m_h5_tk). This cookie acts as a fundamental security token for authenticating subsequent API requests.
- **Step 3: Cryptographic Signature Generation (MD5)**
  - To bypass the platform's API security layer, the system dynamically generates an MD5 hash signature for every request. This signature is constructed by concatenating the extracted security token, the current Unix timestamp, a hardcoded AppKey, and the stringified JSON payload containing the request parameters (e.g., **itemId** and **pageNo**).
- **Step 4: Data Extraction and Pagination**
  - The script iterates through the remaining product IDs. For each product, it sends signed HTTP GET requests to the review API, automatically paginating through the results (handling up to a predefined limit, such as 50 pages or 1000 reviews per item). The parser extracts core variables from the server's JSON response, specifically the *Product ID*, *Star Rating*, *Review Content*, and *Timestamp*.
- **Step 5: Real-time Persistence and Anti-bot Evasion**
  - To mitigate the risk of data loss, the extracted reviews are appended to the local storage immediately after processing each individual product. Furthermore, if the system encounters a rate-limit or an anti-bot block signal (e.g., returning a `FAIL_SYS_USER_VALIDATE` error), it halts the loop, automatically reopens the Selenium browser to allow for manual Captcha resolution, regenerates the security tokens, and seamlessly resumes the crawling process from the exact point of interruption.

2.3 Dataset Construction

After the crawling stage, the raw product and review files were transformed into a relational analytical model for Power BI. This construction step is important because the later dashboard pages do not operate directly on the raw crawler outputs. Instead, the project reorganizes the collected data into dimension tables, fact tables, and an auxiliary segmentation table so that the analytical workflow in Sections 1 and 3 can be supported by a consistent data model.

2.3.1 Construction Logic

- The raw *products.json* and *products.csv* files provide the source product attributes used to construct the product, category, and shop dimensions.
- Consecutive daily product snapshots are aligned by product ID and date in order to form the `fact_product_snapshot` table, which stores the time-varying sales and pricing fields used throughout the dashboard.
- The review crawler output is reshaped into `fact_review`, where each row corresponds to one customer review linked back to the relevant product and date.
- Date attributes are standardized into a dedicated `dim_date` table so that both sales and review facts can share the same analytical calendar.
- Product-level clustering results are exported separately into `product_segments`, then linked back to the conformed product dimension for the machine learning page.

2.3.2 Relational Model Used in Power BI

Based on the implemented Power BI model shown in the relationship view, the final analytical dataset uses the following main tables:

- `dim_category`: category reference table keyed by `category_id`
- `dim_shop`: shop reference table keyed by `shop_id`
- `dim_date`: shared calendar table keyed by `date_key`
- `dim_product`: conformed product dimension keyed by `product_id`
- `fact_product_snapshot`: sales snapshot fact table at the product-date level
- `fact_review`: review fact table at the individual-review level
- `product_segments`: product-level clustering output table

The active relationships visible in the model can be summarized as follows:

- `dim_category(category_id)` $1 \rightarrow *$ `dim_product(category_id)`
- `dim_shop(shop_id)` $1 \rightarrow *$ `dim_product(shop_id)`
- `dim_product(product_id)` $1 \rightarrow *$ `fact_product_snapshot(product_id)`
- `dim_product(product_id)` $1 \rightarrow *$ `fact_review(product_id)`
- `dim_date(date_key)` $1 \rightarrow *$ `fact_product_snapshot(date_key)`
- `dim_date(date_key)` $1 \rightarrow *$ `fact_review(date_key)`
- `dim_product(product_id)` $1 \rightarrow 1$ `product_segments(product_id)`

This structure gives the project a star-schema style core for descriptive analysis, while also attaching the clustering output as a product-level extension. As a result, category analysis, finance structure, product influence analysis, review signals, and clustering insights can all be supported from the same integrated analytical model.

2.4 Data Preparation

This section outlines the data cleaning and feature engineering procedures. Please note that the subsequent Exploratory Data Analysis (EDA) and its visualizations are documented in a separate supplementary file (***eda.ipynb***).

2.4.1 Product Metadata Dataset

This dataset captures the overarching catalog details and aggregate performance metrics for each item in the flagship store. In the raw product metadata snapshot, each row represents one unique product collected during a crawling run.

- **id (Primary Key)**: The unique alphanumeric identifier assigned to the product by the e-commerce platform.
- **ten_san_pham (Product Name)**: The full, descriptive title of the item as displayed on the storefront.
- **danh_muc (Category)**: The specific classification of the product (e.g., specific shoe lines, apparel categories) scraped directly from the store's filter menu.
- **gia_ban (Selling Price)**: The current transactional price of the item, reflecting any active discounts.
- **gia_goc (Original Price)**: The baseline retail price of the item before any promotional deductions.
- **so_luong_da_ban (Sales Volume)**: A quantitative integer representing the historical number of units sold.
- **diem_danh_gia (Rating Score)**: The aggregated average star rating for the product, ranging from 1.0 to 5.0.
- **so_luot_nhan_xet (Review Count)**: The total number of customer ratings the product has accumulated.
- **dia_diem_kho (Warehouse Location)**: The geographical origin or fulfillment center location for the item.
- **ton_kho (Inventory Status)**: A categorical indicator specifying whether the item is currently In-Stock or Out-of-Stock.
- **thuong_hieu (Brand)**: The brand name associated with the item.
- **link_san_pham (Product URL)**: The fully qualified hyperlink to the product's detail page.

2.4.2 Data Cleaning and Standardization For Product Dataset

- **Missing Value Imputation**: Null review counts (so_luot_nhan_xet) were filled with 0, while missing rating scores (diem_danh_gia) were kept as NaN to prevent skewing the mean during statistical analysis.
- **Data Type Casting**: Converted financial and metric columns from text to strict numeric formats (Float/Integer) to enable mathematical calculations.

- **Geographical Standardization**: Merged naming inconsistencies in the warehouse location (dia_diem_kho), specifically mapping "Ho Chi Minh" to the standardized "Thanh pho Ho Chi Minh".
- **Feature Engineering**:
  - **Objective**: Transform cumulative static data (total items sold) into dynamic daily metrics, synchronizing the results directly back to the source files (JSON and CSV).
  - **Baseline Initialization (02-19-2026)**: As the starting point with no prior data, daily sales (so_luong_ban_trong_ngay) for all items are initialized to 0.
  - **Calculation Logic**:
    * Calculated by subtracting the historical maximum recorded in the tracker from the newly scraped cumulative total.
    * Derived by multiplying the newly calculated daily sales by the item's current selling price.

2.4.3 Customer Reviews Dataset

This dataset contains granular, user-generated feedback for the scraped products. Each row represents an individual customer review.

- **id_sp (Foreign Key)**: The product identifier, which maps directly back to the *id* in the Product Metadata dataset, establishing a one-to-many relationship.
- **rating (Star Rating)**: The specific ordinal score (typically 1 to 5) awarded by the individual customer.
- **comment (Review Text)**: The unstructured, qualitative textual feedback provided by the buyer detailing their experience or product quality.
- **time (Timestamp)**: The exact temporal record of when the review was submitted, allowing for time-series sentiment analysis.

2.4.4 Data Cleaning and Standardization For Customer Review Dataset

The extracted review texts and numerical ratings were largely retained in their original state, with only lightweight structural normalization during extraction such as merging multipart review content and trimming text fields where necessary.

3 Data Analysis

3.1 Overview of the Analytical Dataset

The analytical workflow combines three connected evidence layers: product-day sales snapshots, customer review records, and clustering outputs derived from product-level performance features. Instead of relying only on narrative description, this section begins with a compact profile of the prepared analytical dataset and then presents each objective through the dashboard visuals that were actually used in the final report.

Indicator	**Value**
Analytical rows	18,523
Unique products	693
Unique categories	21
Analytical days	29
Average rating score	4.78
Review rows	7,605

Table 3: High-level profile of the analytical dataset

Table 3 shows that the prepared dataset is large enough to support all ten objectives in the report. The sales layer contains **18,523** product-day observations across **693** products and **21** categories over **29** analytical days, while the review layer contributes **7,605** customer review records. The revenue, price, rating, and stock indicators also confirm that the Nike Lazada portfolio is a mid-to-high-priced catalog with generally positive ratings and very limited out-of-stock behavior during the observed period.

`Component`	**Grain / Structure**	**Role in Analysis**
`fact_product_snapshot`	Product-date level	Main sales, price, stock, and rating layer
`fact_review`	Individual review level	Customer feedback and review-signal layer
`dim_product`	Product reference table	Product identity and product attributes
`dim_category`	Category reference table	Category-level grouping and comparison
`dim_date`	Shared calendar table	Time-based aggregation for both facts
`dim_shop`	Shop reference table	Store-level context
`product_segments`	Product-level cluster output	Machine-learning segmentation layer

Table 4: Analytical data structure used in Section 3

From a structural perspective, the analytical dataset follows a star-schema style design centered on two fact tables. The first fact table, `fact_product_snapshot`, supports the sales-oriented objectives because it stores daily observations of revenue, units sold, price, discount rate, stock status, rating score, and review count at the product-date level. The second fact table, `fact_review`, supports customer-feedback analysis because it stores individual review records with rating, comment, and timestamp information. These two facts are connected to shared dimensions for product, category, date, and shop, while `product_segments` extends the model with clustering outputs for the machine learning objectives.

`Variable`	**Observed Distribution Summary**
`selling_price`	Mean = 2.04 million VND; min = 309,000 VND; max = 6.18 million VND
`daily_units_sold`	Mean = 0.19; max = 134; 16,967 zero-sale rows vs 1,556 positive-sale rows
`daily_revenue`	Mean = 320,460 VND per row; max = 164.02 million VND
`rating_score`	Mean = 4.78 across rated observations; ratings available in 15,119 rows
`is_in_stock`	99.55% of product-day rows are in stock
`review_date`	Review rows distributed across 2024–2026, with the largest volume in 2025

Table 5: Distribution summary of the main analytical variables

The variable distributions reveal several important properties of the dataset. First, the selling-price distribution confirms that the portfolio is concentrated in the mid-to-high price range, which is consistent with the finance structure objectives later in the report. Second, the `daily_units_sold` variable is sparse: most product-day rows record zero units sold, while a much smaller number of rows contain positive values and a few extreme peaks. This means the sales pattern is uneven and supports the need for trend and concentration analysis. Third, the `rating_score` and stock variables are both highly favorable overall, with a high mean rating and an in-stock ratio close to 100%. Finally, the review data are not evenly distributed by year, because most of the collected customer-review records fall in 2025, which is important when interpreting the monthly review visuals in Theme 4.

3.2 Theme 1: Sales Performance by Category and Time

3.2.1 Objective 1.1: Total Revenue by Category Name (Column Chart)

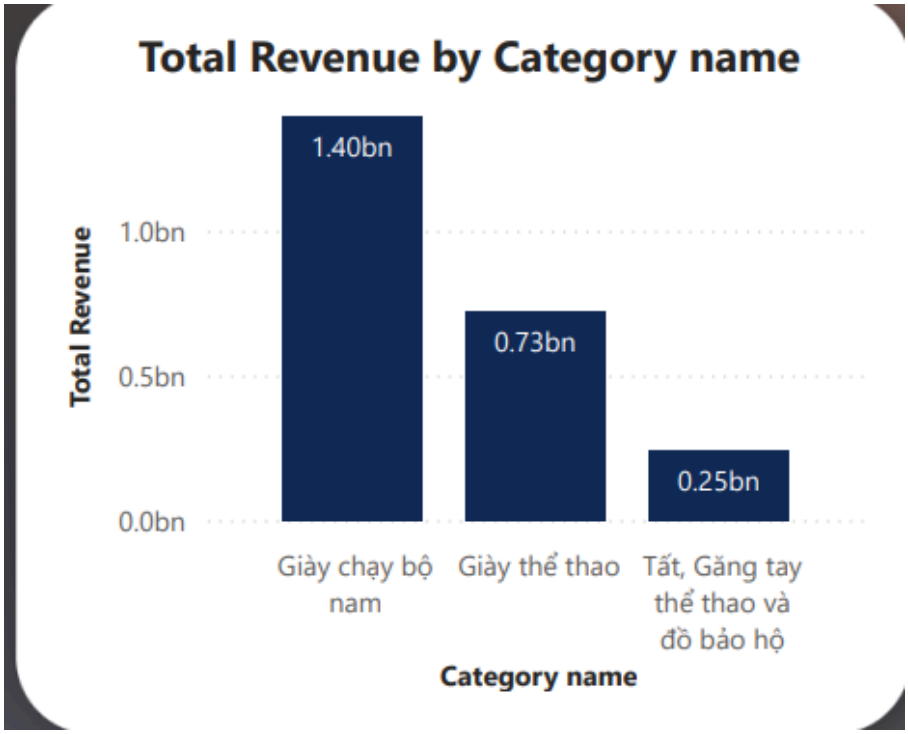
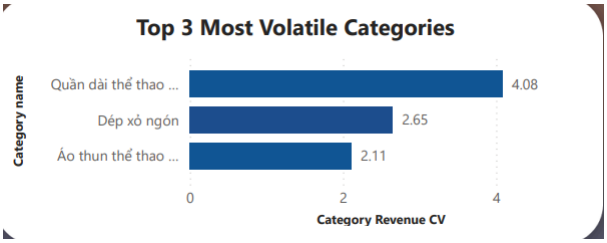


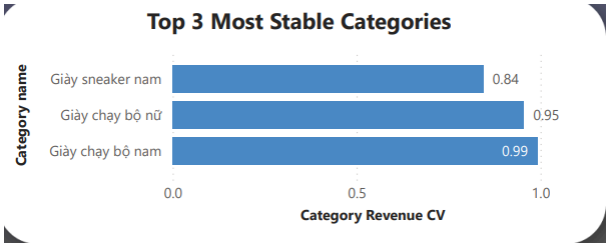
Figure 1: Total revenue by category name

- **Mark:** Vertical bars.
- **Channel:** Bar height encodes total revenue as a quantitative measure, the X-axis position encodes category name as a categorical field, and text labels on bars reinforce exact values.
- **Rationale:** This chart was chosen because Objective 1.1 asks which categories contribute the most total revenue. A column chart answers that objective clearly by placing all categories on a common baseline, allowing direct comparison of revenue magnitude across groups.
- **Insight:** Revenue is clearly concentrated in a small number of categories. *Giay chay bo nam* leads with about **1.40 billion VND**, followed by *Giay the thao* at **0.73 billion VND**, while the third category contributes much less at about **0.25 billion VND**. This indicates that short-term sales performance is driven primarily by a narrow set of core categories rather than by the full catalog evenly.

3.2.2 Objective 1.2: Category Revenue Stability and Volatility (Horizontal Bar Charts)



(a) Top 3 most volatile categories



(b) Top 3 most stable categories

Figure 2: Category revenue CV comparison

- **Mark:** Horizontal bars.
- **Channel:** Bar length encodes `Category Revenue CV` quantitatively, the Y-axis position encodes category names categorically, and value labels show the relative degree of volatility or stability.
- **Rationale:** These charts were chosen because Objective 1.2 asks our group to identify and compare the most volatile and most stable categories. Horizontal bar charts answer that ranking task well by making coefficient-of-variation values easy to order, compare, and interpret side by side.
- **Insight:** The volatility pattern is uneven. The most volatile category reaches a CV of about **4.08**, far above the others, while the stable categories remain below **1.00**. In particular, *Giay sneaker nam*, *Giay chay bo nu*, and *Giay chay bo nam* show relatively consistent performance, whereas categories such as long sports pants and flip-flops depend more heavily on short-lived revenue spikes.

3.3 Theme 2: Finance Structure

3.3.1 Objective 2.1: Price Distribution by Price Band (Pie Chart)

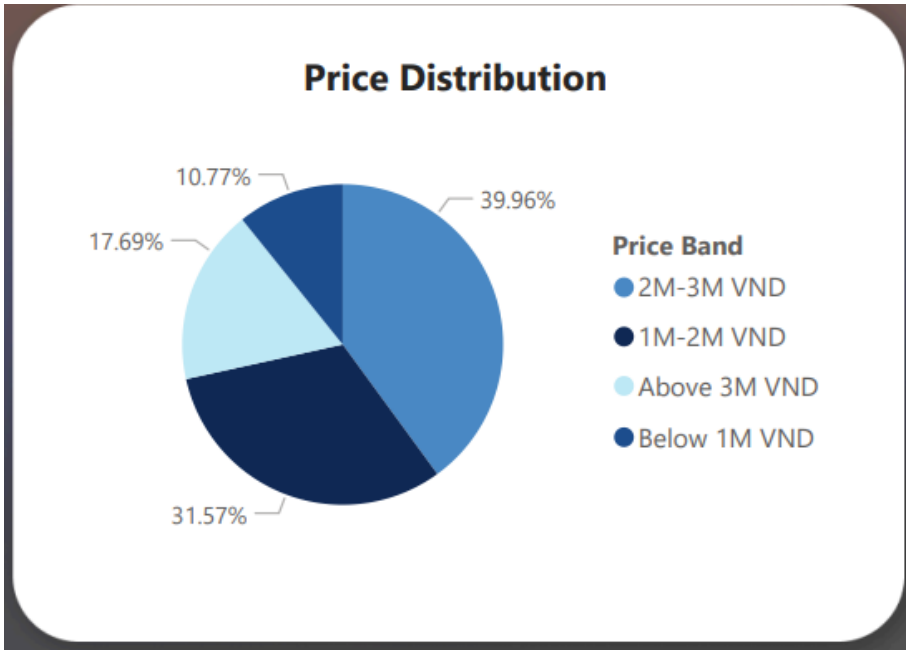


Figure 3: Revenue distribution by price band

- **Mark:** Pie slices.
- **Channel:** Slice angle and area encode revenue share quantitatively, while color distinguishes the four categorical price bands.
- **Rationale:** This chart was chosen because Objective 2.1 focuses on part-to-whole comparison across a small number of price bands. A pie chart directly answers that objective by showing how total revenue is divided among the price groups and by making the dominant price band immediately visible.
- **Insight:** The portfolio's financial center lies in the middle price range. The **2M–3M VND** band contributes the largest share at about **39.96%**, followed by the **1M–2M VND** band at **31.57%**. The **Above 3M VND** and **Below 1M VND** groups contribute much smaller shares, so overall revenue is driven mainly by mid-priced products rather than extreme price segments.

3.3.2 Objective 2.2: Top Trending Daily Revenue Pattern (Line Chart)

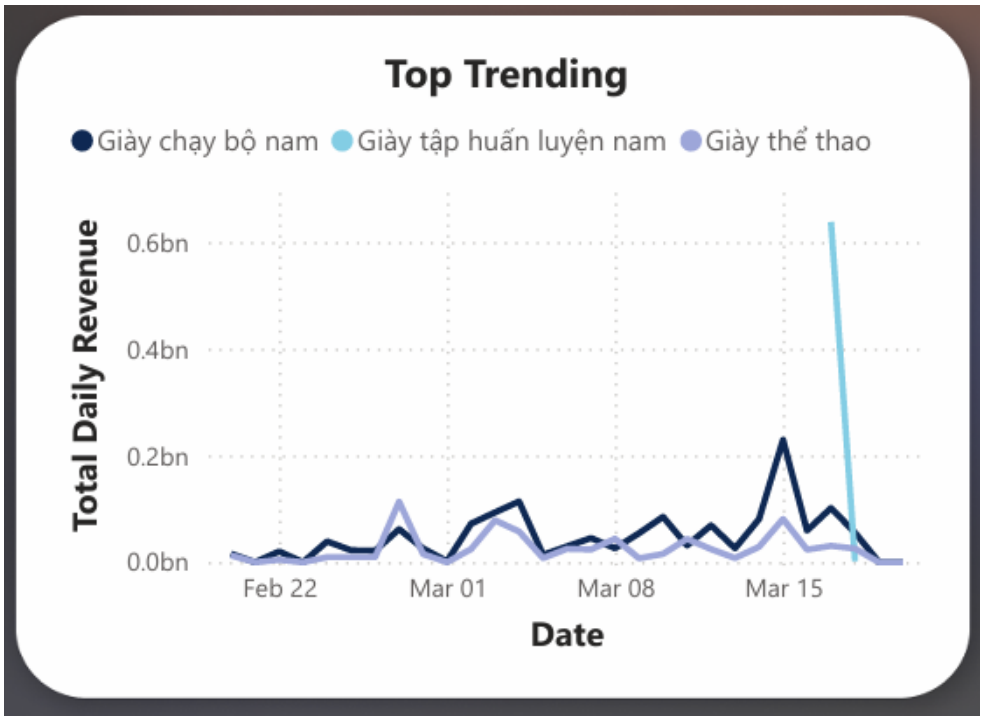


Figure 4: Top trending categories by daily revenue

- **Mark:** Lines.
- **Channel:** Horizontal position encodes date, vertical position encodes total daily revenue, and color differentiates the leading categories shown in the chart.
- **Rationale:** This chart was chosen because Objective 2.2 asks whether revenue behavior is spread across time or concentrated on a small number of dates. A line chart answers that objective effectively by showing daily movement, sudden spikes, and differences in timing across the leading categories.
- **Insight:** Revenue movement is not evenly distributed across the period. Most dates remain at relatively low levels, but there are clear spikes in mid-March, especially for *Giay tap huan luyen nam*, which rises sharply above the other categories. This suggests that page-level financial performance is shaped by a small number of high-impact dates rather than by smooth daily accumulation.

3.4 Theme 3: Product Influence Analysis

3.4.1 Objective 3.1: Top Products by Average Unit Sold (Horizontal Bar Chart)

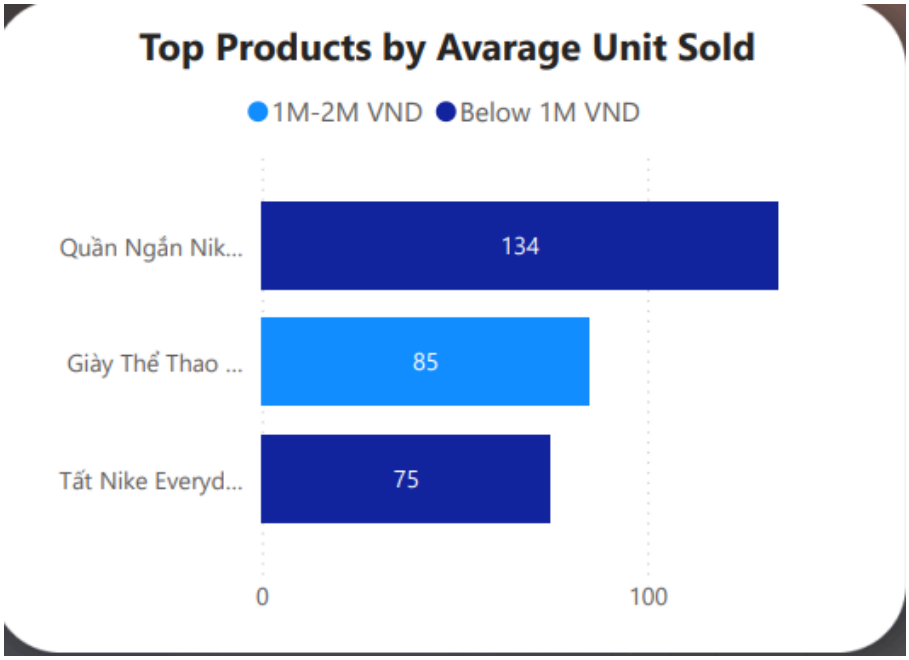


Figure 5: Top products by average unit sold

- **Mark:** Horizontal bars.
- **Channel:** Bar length represents `Avg Daily Units Sold`, Y-axis position represents product identity, and color differentiates the associated price-band grouping shown in the legend.
- **Rationale:** This chart was chosen because Objective 3.1 asks which products lead unit performance. A horizontal bar chart answers that objective clearly by ranking products from strongest to weakest while also giving enough space for long product labels.
- **Insight:** Product influence is concentrated, but not fully monopolized by a single item. The leading product reaches about **134** average units sold, clearly ahead of the second product at **85** and the third at **75**. This shows that a small set of standout items plays an important role in unit performance, although there is still a noticeable spread across multiple top products.

3.4.2 Objective 3.2: Top Product Unit Trend (Line Chart)

Figure 6: Top product unit trend over time

- **Mark:** Lines.
- **Channel:** The X-axis represents date, the Y-axis represents `Avg Daily Units Sold`, and color separates the tracked top products.
- **Rationale:** This chart was chosen because Objective 3.2 asks whether top-product performance is stable over time or driven by a few spike dates. A line chart answers that objective better than a ranking chart because it preserves the time sequence and highlights sudden increases in units sold.
- **Insight:** The strongest products do not perform uniformly across the full period. Instead, the chart shows several sharp spikes in mid-March, including one product peaking above **130** units and another later peak near **85** units. This indicates moderate concentration risk: short-term product influence depends partly on a few strong surge dates rather than on perfectly stable demand.

3.5 Theme 4: Customer Ratings and Review Signals

3.5.1 Objective 4.1: Monthly Rating and Review Trends (Dual-Axis Line Chart)

Figure 7: Monthly rating and review trends

- **Mark:** Two lines on a dual-axis chart.
- **Channel:** The X-axis encodes month, the left Y-axis encodes review count, the right Y-axis encodes average rating, and color distinguishes the two measures.
- **Rationale:** This chart was chosen because Objective 4.1 asks our group to compare two monthly signals at the same time: review count and average rating. A dual-axis line chart answers that objective well by placing both measures on the same timeline while keeping their different numeric scales readable.
- **Insight:** Customer sentiment remains generally high, with the rating line staying in the upper 4.6 to 4.9 range, while review volume varies more noticeably by month. Review activity rises strongly in April and again toward December, whereas average rating dips more clearly around March and November. This means customer attention fluctuates more than customer satisfaction does.

3.5.2 Objective 4.2: Monthly Comment Trends for Top Products (Line Chart)

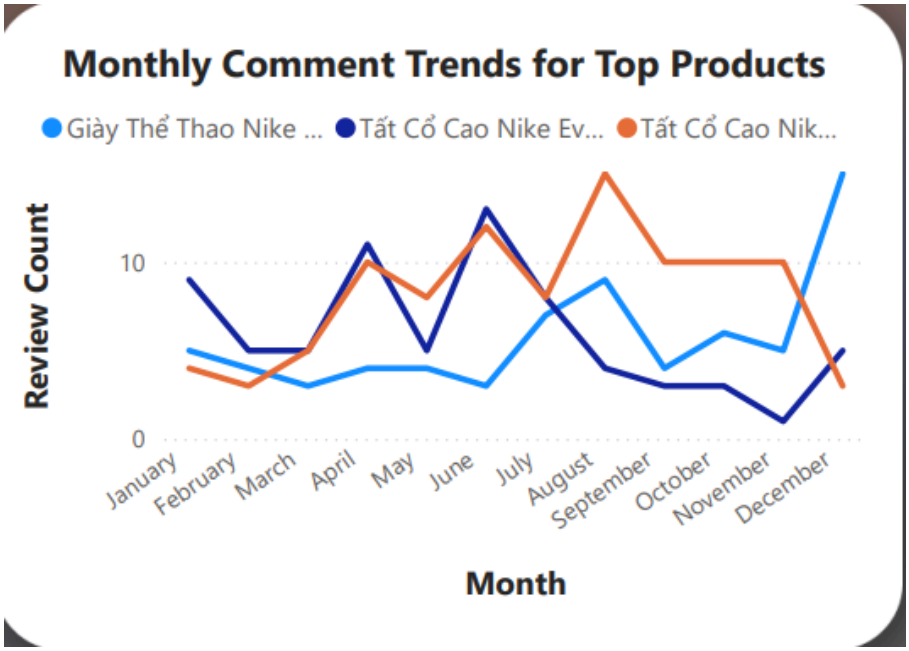
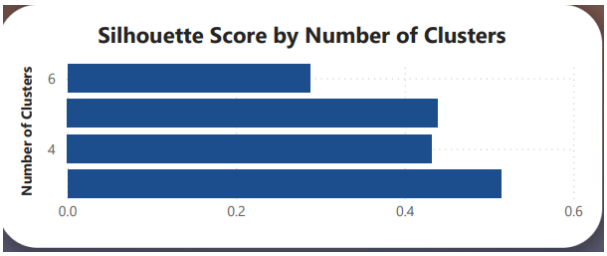


Figure 8: Monthly comment trends for top products

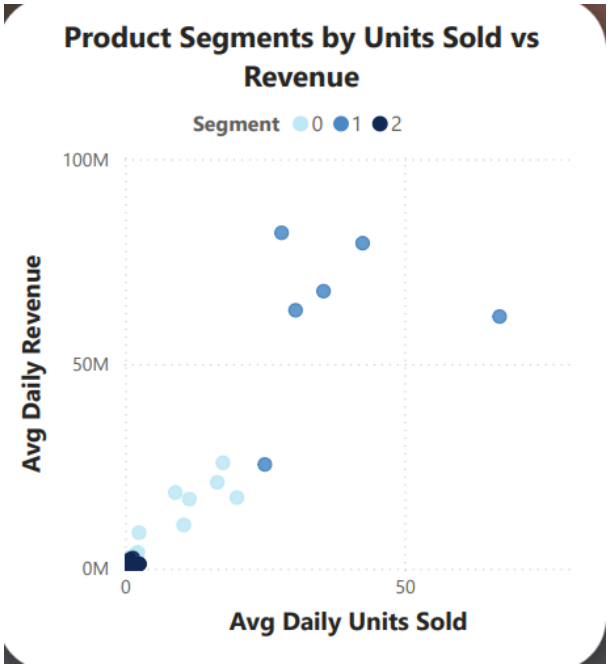
- **Mark:** Lines.
- **Channel:** Horizontal position represents month, vertical position represents review count, and color differentiates the most-discussed products.
- **Rationale:** This chart was chosen because Objective 4.2 asks which products receive the most comments across months and whether that attention stays concentrated. A multi-line chart answers that objective clearly by showing changes in monthly leadership, persistence, and spikes for each tracked product.
- **Insight:** Customer discussion is not dominated by one product throughout the year. Different products lead in different months: one line peaks strongly in August, another becomes more prominent around June, and another jumps highest in December. This suggests that review attention rotates across products rather than remaining permanently concentrated in a single item.

3.6 Theme 5: Machine Learning Insights (Clustering)

3.6.1 Objective 5.1: Clustering Quality and Segment Identification (Bar Chart and Scatter Plot)



(a) Silhouette score by number of clusters

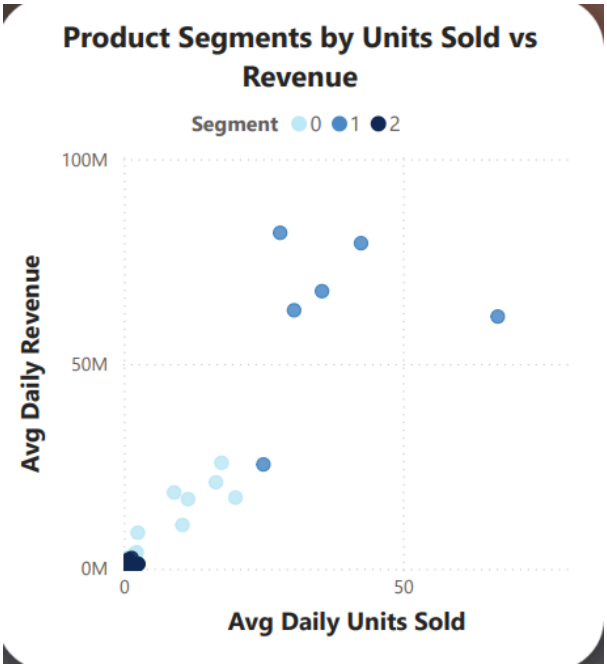


(b) Product segments by units sold vs revenue

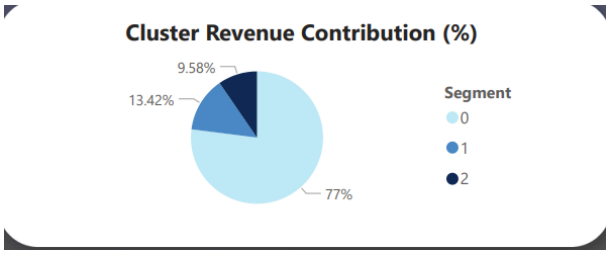
Figure 9: Clustering quality and segment identification

- **Mark:** Horizontal bars in the silhouette chart and points in the scatter plot.
- **Channel:** In the first visual, bar length encodes silhouette score and Y-axis position encodes the candidate number of clusters. In the second visual, X-axis position encodes `Avg Daily Units Sold`, Y-axis position encodes `Avg Daily Revenue`, and color encodes cluster membership.
- **Rationale:** These charts were chosen because Objective 5.1 has two linked tasks: selecting a reasonable clustering solution and checking whether the resulting segments are visibly distinct. The silhouette-score bar chart answers the model-selection part, while the scatter plot answers the segment-identification part by showing whether products separate into different performance regions.
- **Insight:** The **3-cluster** solution appears to be the strongest because it has the highest silhouette score, around **0.53**, while the alternatives with 4, 5, and 6 clusters perform worse. The scatter plot also shows meaningful separation between low-performing products near the origin, mainstream products in the middle range, and a smaller high-performing group with much larger units sold and revenue values.

3.6.2 Objective 5.2: Cluster Revenue Contribution Interpretation (Scatter Plot and Pie Chart)



(a) Product segments by units sold vs revenue



(b) Cluster revenue contribution

Figure 10: Cluster contribution to overall sales performance

- **Mark:** Points in the scatter plot and pie slices in the contribution chart.
- **Channel:** The scatter plot uses spatial position to compare performance by units sold and daily revenue, while the pie chart uses slice area and color to show each segment's share of total revenue.
- **Rationale:** These charts were chosen because Objective 5.2 asks not only how clusters differ, but also how much each cluster contributes to overall sales performance. The scatter plot explains the behavioral profile of each segment, while the pie chart directly answers the contribution question by showing revenue share across clusters.
- **Insight:** Segment **0** contributes the largest share of total revenue at about **77%**, so the mainstream product group remains the main commercial base of the portfolio. The remaining segments contribute about **13.42%** and **9.58%**, which shows that smaller groups still matter but play more specialized roles. Combined with the scatter plot, this indicates that overall revenue is dominated by the broad central segment rather than by the extreme high-performing or low-performing groups alone.

3.7 Overall Summary of Findings

The objective-based visual analysis supports the group's common analytical problem in a coherent way. Category revenue is concentrated in a few strong groups, the portfolio's financial center lies in the 1M–3M VND range, and product influence is shaped by several standout items with visible mid-March spikes. Customer review volume changes more

strongly than average rating, while the clustering layer shows that the catalog can be meaningfully separated into segments with different performance profiles and different revenue contributions.

Overall, the report now aligns the analysis narrative directly with the dashboard evidence used in the final submission. This makes Section 3 more consistent with the actual visuals, easier to defend in presentation, and more clearly connected to the measurable objectives introduced in Section 1.

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